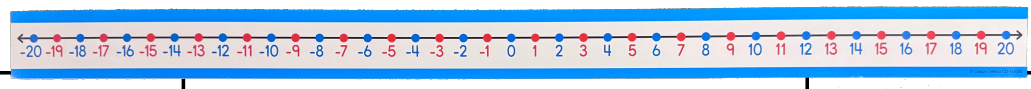
U1A L1 Solving 1 Step Equations **NOTES** Name: _____HR:_____

Target Skill: Solve the 4 types of 1 step equations and name the step.

Prerequisite skills:



Additive Inverses

Given a real number "a" there is always another number "-a" that is its opposite such that:

$$a + (-a) = 0$$

3 and -3, -8 and 8, -5.2 and 5.2

$3 + -3 = 0$ $-8 + 8 = 0$ $-5.2 + 5.2 = 0$

Addition/Subtraction

$$\begin{array}{r} -17 \\ +12 \\ \hline -5 \end{array} \qquad \begin{array}{r} 23 \\ -28 \\ \hline -5 \end{array}$$

$17 - 12 = 5$ $28 - 23 = 5$

$17 + 12 = 29$ $-17 - 12 = -29$

$\frac{2}{3} \div 2 \div 3$

$\frac{2}{3} \cdot \frac{5}{5} = \frac{10}{15}$ $+ \frac{3}{5} \cdot \frac{3}{3} = \frac{9}{15}$ $= \frac{19}{15}$

$\frac{-5}{8} \cdot \frac{2}{2} = \frac{-10}{16}$ $+ \frac{3}{4} \cdot \frac{4}{4} = \frac{12}{16}$ $\frac{-5}{8} = -\frac{5}{8}$ $\frac{3}{4} \cdot \frac{2}{2} = \frac{6}{8}$ $= \frac{1}{8}$

$\frac{2 \div 2}{16 \div 2} = \frac{1}{8}$

Multiplication (with Fractions

$\frac{2}{3} \cdot \frac{4}{5} = \frac{8}{15}$

$2 \cdot \frac{3}{8} = \frac{6}{8} = \frac{2 \cdot 3}{2 \cdot 4} = \frac{3}{4}$ $12 \cdot \frac{3}{8} = \frac{3}{4}$

$\frac{6}{1} \cdot \frac{1}{6} = \frac{6}{6} = 1$

$\frac{3}{5} \cdot \frac{5}{3} = \frac{15}{15} = 1$

$x \cdot \frac{1}{x} = 1$

multiplicative inverse

$\frac{a}{b}$ & $\frac{b}{a}$

Objective: Solve equations by getting the variable alone.

Rules of the math game:

1. **You can add or subtract the same number from BOTH SIDES. (Addition/Subtraction Property of Equality)**
2. **You can multiply or divide both SIDES by the same number (Multiplication/Division Property of Equality**
3. **You can exchange the left side and the right side for each other (Symmetric Property)**

$a = b \leftrightarrow b = a$

Tools of the Game:

- **Additive inverses:** $\begin{array}{r} a \\ -a \\ \hline 0 \end{array}$ **or** $\begin{array}{r} -a \\ +a \\ \hline 0 \end{array}$
- **Multiplicative Inverses/Reciprocals:**

$\frac{5}{1} \cdot \frac{1}{5} = 1$ $\frac{2}{3} \cdot \frac{3}{2} = \frac{6}{6} = 1$

Directions: Determine the value of x that makes the equation true. STATE the RULE(s) used. Answer all as $x =$ answer

1. $x - 3 = 5$

$+3 \quad +3$

$x = 8$

$x = $ 8

Property Used: Add. Prop. of =

2. $-2 = x - 8$

$+8 \quad +8$

$6 = x$

$8 - 2 = 6$

$x = $ 6

Property Used: Add. prop of = Symmetry

3. $x - \frac{5}{3} = \frac{7}{6}$

$+\frac{5}{3} \quad +\frac{5}{3}$

$x = $ 17/6

Property Used: Add. prop. of =

INVERSE OPERATIONS: Add and Subtract are called INVERSE OPERATIONS. They "UNDO" each other.

The inverse of ADD 3 is SUBTRACT 3 or the inverse of SUBTRACT 3 is ADD 3.

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4. $x + 4 = 9$

$-4 \quad -4$

$x = 5$

$x =$ 5

Property Used: Sub. prop. of =

5. $-1 = 6 + x$

$-6 \quad -6$

$-7 = x$

$x = -7$

$x =$ -7

Property Used: Sub. prop of = symmetric prop

6. $x + \frac{2}{5} = \frac{9}{10}$

$-\frac{2}{5} \quad -\frac{2}{5} \cdot \frac{2}{2}$

$\frac{9}{10} - \frac{4}{10} = \frac{5}{10} = \frac{1}{2}$

$x = \frac{1}{2}$

Property Used: Sub. prop. of Eq

The next problems, the operations done to x have changed to Multiplication and division this can be confusion sometimes to determine if you should think of it as a multiplication done to x or a division.

- **In the equation:** $\frac{x}{5} = 2$ **what does the equation show is being done to** x**?**

DIVISION

$\frac{x}{5} = 2$

$x \div$ by 5

- **In the equation:** $5x = 2$ **what does the equation show is being done to** x**?**

$5 \cdot x = 2$

$5x$

"5 times x"

- **In the equation:** $\frac{5x}{3} = 2$ **what does the equation show is being done to** x**?**

$\frac{5x}{3} = 2 \quad \frac{5}{3} \cdot x$

"x times a 5 & ÷ by a 3" OR "x time a $\frac{5}{3}$"

7. $\frac{x}{4} = 3$ ·4

$x = 12$

$x =$ 12

Property Used: mult. prop of =

8. $\frac{-2 \cdot (-5)}{1} = \frac{x}{-2} \cdot \frac{-2}{1}$

$10 = x$

$x =$ 10

Property Used: • Mult. prop of = • Symmetric

9. $\frac{x}{7} = \frac{5}{2} \cdot \frac{7}{1}$

$x = \frac{35}{2}$

$x =$ $\frac{35}{2}$

Property Used: Mult. prop of =

10. $5x = 20$

$\frac{5x}{5} = \frac{20}{5}$

$x = 4$

$x =$ 4

Property Used: Division prop of =

11. $6 = -2x$

$\frac{6}{-2} = \frac{-2x}{-2}$

$\frac{-2}{-2} = 1$

$-3 = x$

$x =$ -3

Property Used: Div. prop of = Symmetry

12. $\frac{2}{3}x = 5$

$\frac{2}{3} \quad \frac{2}{3}$

$x = \frac{15}{2}$

$x =$ $\frac{15}{2}$

Property Used:

$\frac{3}{1} \cdot \frac{2}{3} x = 5 \cdot 3$

$\frac{2x}{2} = \frac{15}{2}$

$x = \frac{15}{2}$